

Simplified Course Registration: Improving Timetable Building for Carleton University Students

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Abstract

I investigated the need to improve course registration for Carleton University students by comparing a Figma mockup interface with the original registration method interface from Carleton Central. In the study, I surveyed an online within-subjects experiment involving six participants (all students, ages 20-23). The task in the experiment involved registering for any two courses on both interfaces during the Discord call. I set the independent variable as the interface type: the original interface versus the Figma interface. I measured three dependent variables: registration errors, task completion time, and NASA-TLX workload survey scores. I tested the following three hypotheses:

1. Figma leads to no difference (H_0) versus fewer errors (H_1) than Carleton Central.
2. There exists no correlation (H_0) versus a correlation (H_1) between registration time and workload.
3. Figma causes no difference (H_0) versus a decrease (H_1) in registration time compared to Carleton Central.

I rejected the null hypotheses in (1) and (3) because the data supported both fewer errors and lower registration time using the Figma mockup compared to Carleton Central. However, I did not reject the null hypothesis in (2), because there was no correlation between the two based on the results. Participants universally praised the benefit of a side-by-side view of course search and timetable with live interaction and feedback in the Figma mockup interface. In contrast, they found Carleton Central lacking these features and was generally harder to navigate. A few limitations of the study include a small sample size of students, Limitations include my small sample size, the choice of only two courses, and the one-day experiment duration (under an hour). I suggest future research include graduate students, testing multiple Figma interfaces that differ in style, and more courses.

Introduction, Goal, and Hypotheses

Carleton Central is an administrative and academic website for students enrolled at Carleton University, as shown in both **Figure A** and **Figure B**. I tasked students with registering for courses in an upcoming semester for the experiment, shown in the **Figure C** screenshot. This involved clicking Build Your Timetable/Registration in the picture. I noticed that Carleton Central's course registration system provides unclear feedback or lacks it entirely (Assignment 3), forces excessive page-switching (Assignment 3, Question 2), and features an outdated interface. These problems violate Nielsen's principles of visibility and minimalist design (Assignment 3, Lecture 10), increase cognitive load (Lecture 4), and heighten student stress and frustration during course registration. With these issues in mind, I designed my Figma mockup to try and resolve all of these issues. The design features a side-by-side layout with visible action feedback. I incorporated feedback to allow recognition rather than recall (part of Nielsen's design

principles). The bigger picture of the experiment was to compare this prototype against the original interface, showing the differences between the efficiency and user experience.

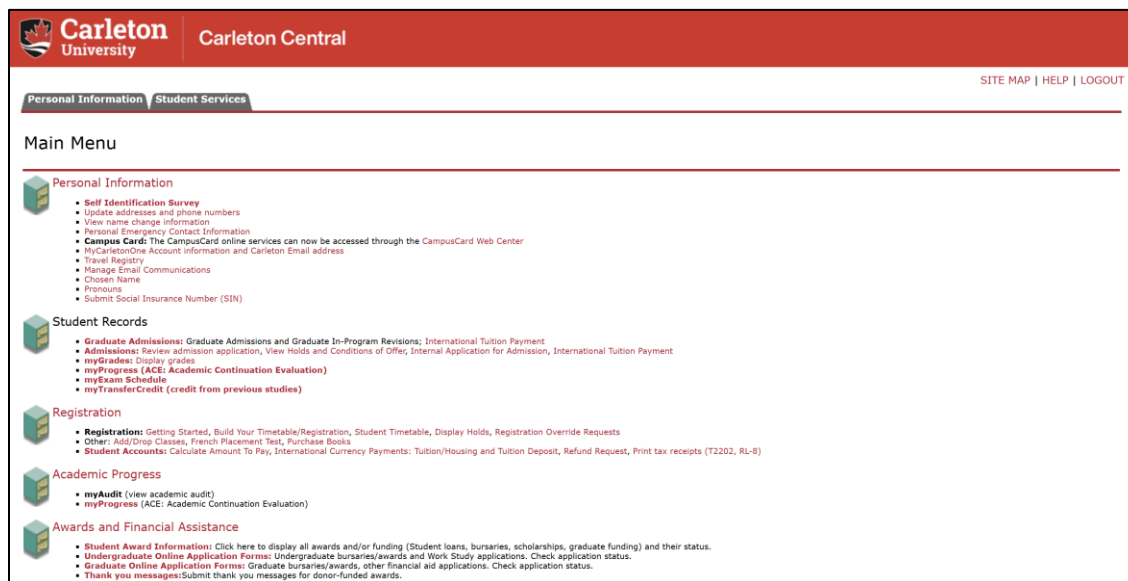


Figure A – First half of the home page

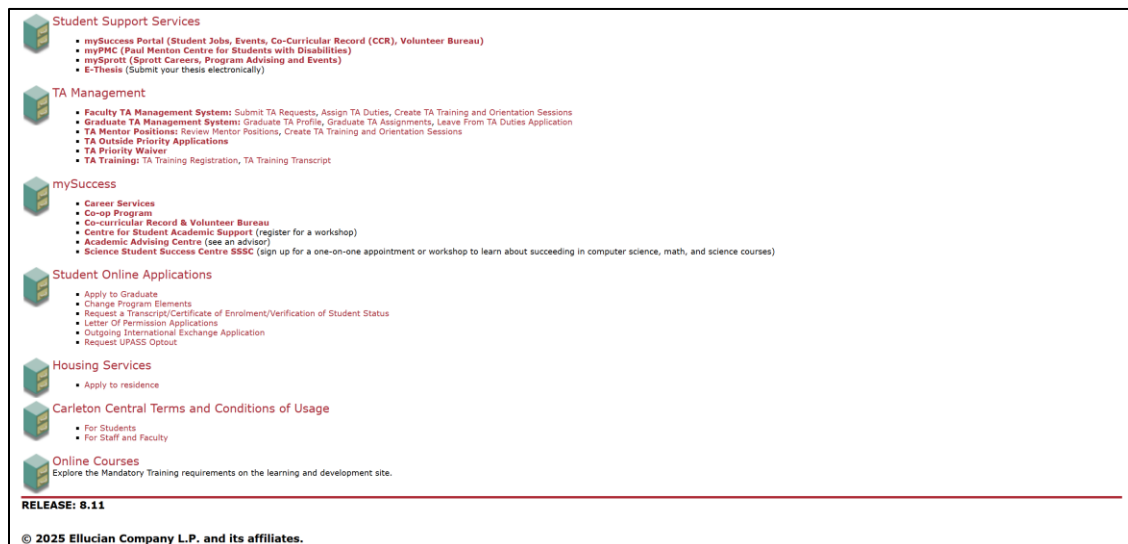


Figure B - Second half of the home page

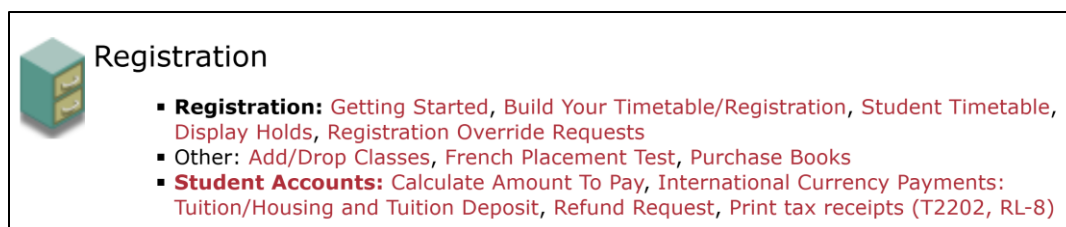


Figure C – Registration options on the Carleton Central site

Below are the three hypotheses related in the experiment, where null is H_0 and alternative is H_1 :

1. H_0 : I observe no difference in error rates between the original Carleton Central and Figma mockup interfaces.
 H_1 : I measure fewer errors with the Figma mockup.
2. H_0 : I find no correlation between task completion time and NASA-TLX scores for either interface.
 H_1 : I detect a linear relationship between task completion time and NASA-TLX scores.
3. H_0 : I observe no difference in task completion time between Original Carleton Central and Figma mockup interfaces.
 H_1 : I measure lower task completion time using the Figma mockup.

While analyzing the experiment results, I either accept or reject the null hypothesis for each of the hypotheses above.

Methods

The study involved six Carleton University students, a with an average age of 21.33 years. I set inclusion criteria requiring participants to be current students at Carleton University, aged 18-30, familiar with Carleton Central (from prior registration experience), and owning a personal computer or laptop. I chose broad criteria as most Carleton students meet them, and it allowed me to find participants easily. For the experiment, I relied on convenience sampling, the easiest method for me, selecting a group of people from my friends and acquaintances. I invited participants via text or social media with brief details and explanations. Despite my efforts, this approach introduced potential bias since I knew each participant, and they might have wanted to please me with their answers. The sample of six students did not properly represent the large Carleton student population. Despite this, it was a good starting point to learn about the differences between the Figma mockup and the original Carleton Central registration. I conducted the experiment online via Discord video calls, the most straightforward method, allowing students to participate comfortably from home. Additionally, the remote setup mirrored real-world conditions as students register for courses at home on their computer. Participants used their own laptops or desktops, with a keyboard and mouse. began each session by saying "Start," then used a smartphone stopwatch to record task completion times precisely to the nearest whole second. I assumed participants were already logged into Carleton Central and that the Figma mockup functioned as expected. During sessions, I required participants to turn on their webcams and microphones and share their screens, enabling me to observe their behavior and screens live. I did not record audio or video, but I took notes and counted the errors each student made. I employed a within-subjects design (Lecture 17); each participant tested all levels of the independent variable. Specifically, I had all six participants test both the Original Carleton Central registration and the Figma mockup in the same session together. This approach allowed

me to directly compare their performances using each interface. I defined the independent variable (interface type) with two levels: Carleton Central (current website) and Figma (proposed redesign). I measured three dependent variables:

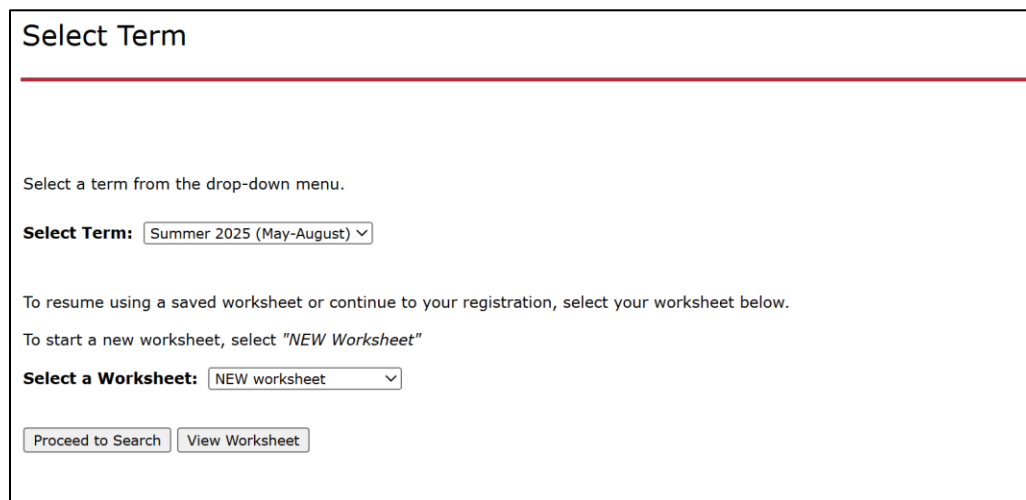
- Task Completion Time (continuous): Time (in seconds) from the home page to successful submission of two courses in course registration. The time was recorded by my phone.
- Number of Errors (categorical): I counted the numeric total of mistakes, such as clicking the wrong course or adding incorrect courses, while watching the students.
- NASA-TLX Workload (subjective): calculated a score (0-100) from six subscales: Mental Demand, Physical Demand (reverse-coded), Temporal Demand (reverse-coded), Performance, Effort, and Frustration. There was a survey for this on Qualtrics following each interface trial.

I scheduled each participant's trials at separate times. The NASA-TLX survey provided me with a measure of cognitive and emotional demands, complementing the objective metrics.

The procedure structure ensured consistency and reliability for every participant. To begin, each participant was asked to complete the consent form on Qualtrics, as required. I designed the form to help them understand how the experiment worked and their rights and responsibilities. I explained everything in the form, including how I would use their data, and ensured no participant felt forced to agree. If they choose "Yes", it indicates their interest in participation. Consent is mandatory to participate in the experiment; those who declined are excluded, though none of the participants declined. Next, I called each student one-on-one through Discord and briefly reviewed the experiment again to ensure clarity and answer questions. Once they had no more questions, I asked each participant to log in and navigate to the registration menu in Carleton Central, as shown in **Figure D**. I made the participants turn on their webcam and microphones and share their computer/laptop screen in preparation to start the experiment. This was necessary for me to monitor their interaction, and time their progress without recording anything for later use. Once everything was completely set up (login, webcam, microphone, shared screen, clear instructions), I conducted a test run to verify functionality and understanding. I informed participants ahead of time that the courses they chose did not matter—just any two of their preference. After a successful test run, I began the real experiment. The participants started with the original Carleton Central website for the first trial, and then Figma mockup interface for the second trial.

Each participant chose the two courses they registered for in the Summer 2025 term. I determined the number of courses, as most students typically only register for a few summer courses and not a full semester (unlike in Fall or Winter semesters). The courses were different from the ones in the test run to prevent reliance on memory. Once ready, I gave the indication to begin and started the timer on my phone. During the test, I watched the students and noted down errors and general behavior of the participants. Examples of errors I tracked included choosing

the wrong course, changing their mind on a selected course, picking an ineligible course (due to restrictions), or encountering course conflicts. Once they reached the registration button (without registering, just hypothetically), I stopped the timer. I did not want the students to register for courses they would not take. Following the first trial, the student(s) completed the various questions on the NASA-TLX survey on Qualtrics. This survey let them rate their workload across six criteria: Mental Demand, Physical Demand, Temporal Demand, Performance, Effort, and Frustration. After this, we talked a little bit about how the first trial went and their NASA-TLX survey results. In total, this trial, from the consent form to the discussion of the NASA-TLX survey result took around 20-30 minutes. After this, there was a 5-to-10-minute break in between. After the break, the next trial on Figma began. The students again only start when I gave the green light to begin. On Figma, each participant registered for two different courses. Timing, error tracking, and NASA-TLX completion followed are the same from the first trial but are now based on the Figma mockup interface. This trial was shorter since the consent form was already filled out. In total, this trial took somewhere between 15-20 minutes. Once complete, the participants filled out the post-experiment questionnaire on Qualtrics. The students provided demographic data (age), ratings of each interface, and qualitative feedback on preferences and clarity. The session concluded with a debriefing, where I thanked each participant, explained the study's aim to improve Carleton Central, and allowed any final questions they had for me or about the experiment. In total, the entire experiment for each participant took no more than 1 hour. Following the experiment, all data was made anonymous with labels P1 to P6 (P = Participant) to protect each participant's identity and was ready for use in the analysis/results.



Select Term

Select a term from the drop-down menu.

Select Term: Summer 2025 (May-August) ▾

To resume using a saved worksheet or continue to your registration, select your worksheet below.

To start a new worksheet, select "NEW Worksheet"

Select a Worksheet: NEW worksheet ▾

[Proceed to Search](#) [View Worksheet](#)

Figure D – Registration Menu on the original Carleton Central website

Results

I analyzed the results with the help of Python. In the file called analysis.py, I tested my hypotheses with a paired t-test for time, Pearson correlation for time-NASA-TLX, and

descriptive statistics for errors, along with other statistics. The file imported four libraries: pandas, scipy.stats, numpy, and matplotlib, each having its own purpose.

- pandas: This is a Python library used to manipulate and analyze data.^[1] I used it to create and manage the dataset (e.g., storing participant data in a table) and filtering the data of each interface.
- scipy.stats: It has statistical functions for hypothesis testing and analysis.^[2] I used this library for the paired t-test and Pearson correlation, which helped me to evaluate my hypotheses using statistics.
- numpy: Importing this library allows you to handle different types of arrays and quickly calculate different mathematical computations in Python.^[3] I used it to create arrays and calculate values, including means (averages) and standard deviations.
- matplotlib: Using matplotlib helps plot data and create visualizations like graphs and charts in Python.^[4] I used it to generate all of the graphs to visualize the results.

Using these libraries made it much easier to present the data. Each of the hypotheses were looked at to determine whether we reject the null hypothesis or not. For reference, here is the data that was used in the calculations. The data is in order as is from P1 to P6, where the letter P represents participant. Note that numbers were rounded to two decimal places in most cases (calculations and data, for simplicity).

Time (s):

- Original Carleton Central [183, 200, 199, 220, 180, 250]
- Figma [76, 90, 88, 79, 55, 103]

Errors:

- Original [2, 2, 4, 3, 2, 1]
- Figma [0, 0, 0, 1, 0, 0]

NASA-TLX:

- Original Carleton Central [65.83, 61.67, 71.67, 71.67, 58.33, 75.83]
- Figma [33.33, 36.67, 38.33, 40.00, 25.83, 45.00]

Ages: [22, 21, 21, 20, 23, 21]

Hypothesis 1 - Error Rates

I tested H_0 (I observe no difference in error rates between the original Carleton Central and Figma mockup interfaces) versus H_1 (I measure fewer errors with the Figma mockup). The mean was calculated in Python by summing each value and dividing by the number of values. The standard deviation was calculated in Python using the formula below in **Figure E**.

$$s.d. = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}}$$

where x_i represents each datapoint $(x_1, x_2, \dots, x_n)$
 $\bar{x}$ is the mean,
 n is the number of values.

Figure E - standard deviation formula [5]

The results are below in **Table 1**.

Table 1 – Data for Hypothesis 1

	Original Carleton Central	Figma
Sum of Errors	14	1
Mean	rounded to 2.33	rounded to 0.17
Standard Deviation	rounded to 1.03	rounded to 0.41

Figma had more than 13 times fewer errors on average than the original Carleton Central, which is a very large margin. Also, there are no outliers in either of these categories, meaning that these results are quite consistent. Therefore, I rejected H_0 and accepted H_1 , as there was undeniable proof that the Figma interface had less errors. The reason is likely due to the simplicity of the Figma interface. Also, they were able to see the timetable and course search side by side, so there were few errors that could possibly occur.

Hypothesis 2: NASA-TLX Correlation

In the second hypothesis, I tested H_0 (I find no correlation between task completion time and NASA-TLX scores for either interface) versus H_1 (I detect a linear relationship between task completion time and NASA-TLX scores). To calculate the Pearson Correlation, Python used the formula that matches the one below in **Figure F**.

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

Where,
 r = Pearson Correlation Coefficient
 x_i = x variable samples y_i = y variable sample
 $\bar{x}$ = mean of values in x variable $\bar{y}$ = mean of values in y variable

Figure F – Formula to calculate the Pearson Correlation Coefficient [6]

To get the p-value to test the null hypothesis, the Python code uses both the t-statistic formula and the probability density function, found in **Figure G** and **Figure H**.

Formula

$$t = \frac{m - \mu}{s / \sqrt{n}}$$

t = Student's t-test
 m = mean
 μ = theoretical value
 s = standard deviation
 n = variable set size

Figure G – T-statistic formula used below ^[6]

$$f(x, \nu) = \frac{\Gamma((\nu + 1)/2)}{\sqrt{\pi\nu}\Gamma(\nu/2)} (1 + x^2/\nu)^{-(\nu+1)/2}$$

where x is a real number and the degrees of freedom parameter ν (denoted `df` in the implementation) satisfies $\nu > 0$. Γ is the gamma function (`scipy.special.gamma`).

Figure H – Formula for the probability density function ^[7]

The results are below in **Table 2**.

Table 2 – Data for Hypothesis 2

	Original Carleton Central	Figma
Pearson correlation coefficient (r)	0.12	0.11
p-value	rounded to 0.83	rounded to 0.85

From the lecture, it showed that a Pearson Correlation coefficient between 0.1 and 0.3 meant that a small correlation existed. This was true for both the Original Carleton Central and Figma interfaces, with very similar values. The p-value here was much greater than 0.05 for both, meaning that that deviation from the null hypothesis is not statistically significant. ^[8]. Therefore, I did not reject the null hypothesis. There is a lack of evidence from the experiment to disprove that there was no correlation between task completion time and NASA-TLX scores.

Hypothesis 3: Task Completion Time

Finally, I tested the third hypothesis: H_0 (I observe no difference in task completion time between Original Carleton Central and Figma mockup interfaces) versus H_1 (I measure lower task

completion time using the Figma mockup). For this test, I used Python to calculate the mean, standard deviation, t-statistic (**Figure G**) and p-value for the task completion time for each interface. Each of these formulas were used in the earlier two hypotheses, which is the same process that Python used. The results are below in **Table 3**.

Table 3 - Data for Hypothesis 3

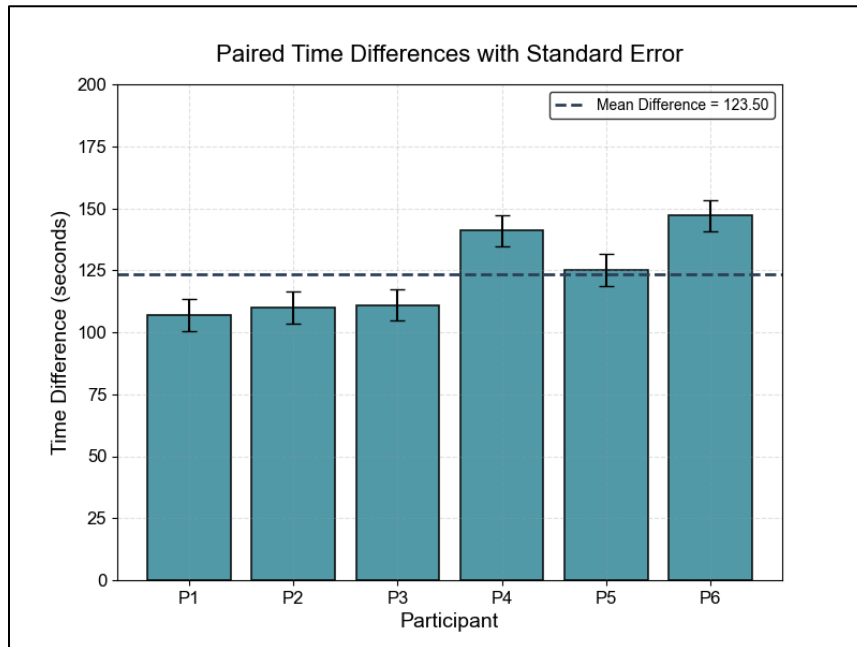
	Original Carleton Central	Figma
Sum of Completion time	1232	491
Mean	rounded to 205.33	rounded to 81.83
Standard deviation	26.41	16.28

Paired t-test results: t-statistic = 10.99 and p-value < 0.01

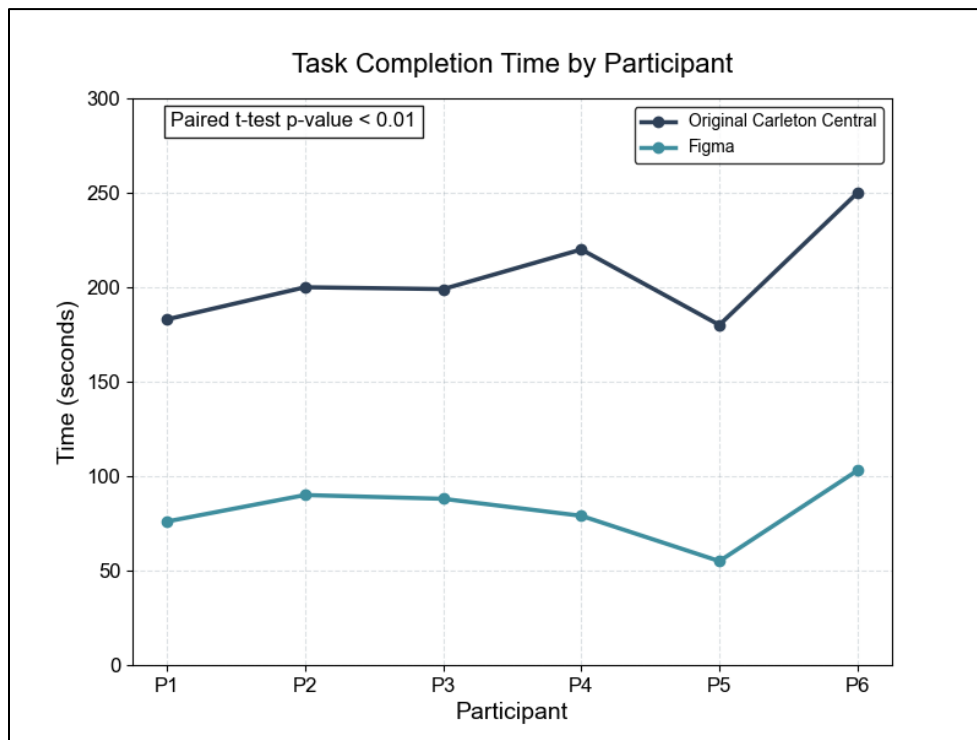
To compare the task completion times, I did a paired t-test using the Python function `stats.ttest_rel` ^[9] from the `scipy.stats` library. I chose to do paired t-test ^[10] rather than an independent t-test since the experiment used a within-subjects design. Each of the participants completed their registration on both interfaces. This means the same individuals provided data for both conditions. The paired t-test accounts for this dependency by analyzing the differences in their performance across the two interfaces. However, the independent t-test compares two separate groups, which is not applicable to this experiment. To begin, Python calculated the mean difference between the paired observations (subtracting the averages), then verify if this difference is statistically significant using the t-statistic and p-value ^[11]. Here was each part broken down.

Differences: For each participant, I (using Python) subtracted the Figma time from the Original time, which resulted in the following times: 107, 110, 111, 141, 125, 147. Taking the mean of this when added together and dividing by 6, you get the mean difference ($\bar{d}$) = 123.50. Next, I calculated the Standard Deviation of Differences using the formula above for standard deviation. First, this involved finding the sum of the square differences: $(107 - 123.50)^2 + (110 - 123.50)^2 + (111 - 123.50)^2 + (141 - 123.50)^2 + (125 - 123.50)^2 + (147 - 123.50)^2 = 1471.50$, then I took this value and plugged it into the standard deviation formula. Python's formula differs from the one in **Figure G**, rather taking the square root of the square differences divided by the sample size - 1: $s_d = \sqrt{\frac{\text{sum of square differences}}{n-1}}$, which gives you $s_d \approx 17.16$. Then, the t-statistic is calculated as $t = \frac{\bar{d}}{s_d/\sqrt{n}}$. This calculated the t-statistic to be 10.99. Lastly, I used the Python `stats.ttest_rel` function to compute the two-tailed p-value, yielding $p = 0.000104$. We can confidently say that $p < 0.01$. The t-statistic indicated a large difference relative to the variability in the data, and the p-value is much lower than 0.05. Therefore, I reject

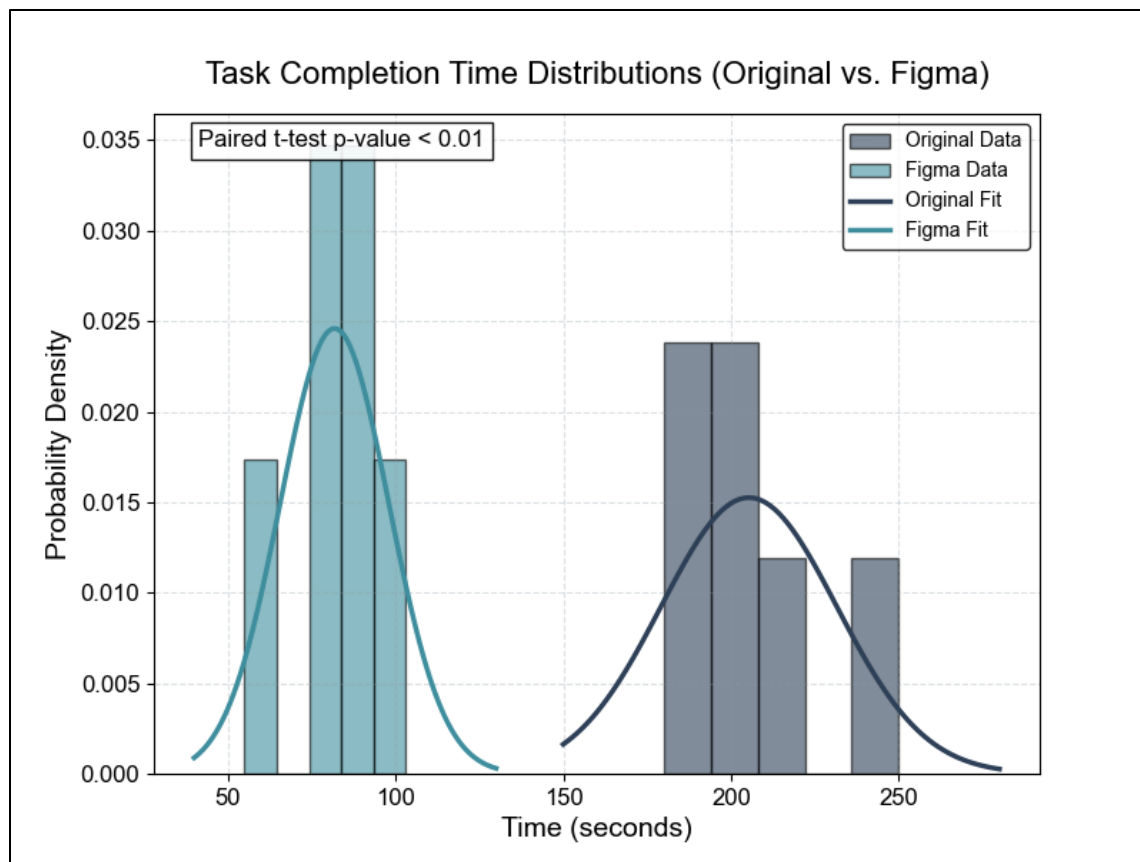
H_0 and accept H_1 . This confirms that the Figma mockup significantly reduced task completion time compared to the Original Carleton Central interface. Here are two graphs, labelled **Graph A**, **Graph B** and **Graph C** to show the data from the paired t-test visually.



Graph A – Difference of participants using standard error, based on completion time



Graph B – Completion time of registration from students in both interfaces



Graph C – Completion time comparison between both interfaces

Discussion

In the study, I was able to gain some insightful knowledge on how some students experience Carleton Central and the Figma mockup through a sample of 6 Carleton University students. The Figma mockup interface seemed to be largely favoured over the original Carleton Central. Looking at Hypothesis 1, I observed a dramatic drop in errors from an average of 2.33 with the Original Carleton Central interface to 0.17 with my Figma redesign, supporting H_1 . Clearer feedback in Figma likely minimized slips, which was a topic discussed in our course (Lecture 8). Hypothesis 2 yielded no correlation between task completion time and NASA-TLX scores; there was no obvious linear relationship. This led me not to reject H_0 . For Hypothesis 3, I conducted a paired t-test and rejected H_0 , because I measured lower task completion time using the Figma mockup. The task completion time declined drastically, and the NASA-TLX scores dropped from by over 50% on average, indicating lower perceived workload. All participants praised how Figma's had the split-screen viewing for the course search and timetable, as well as and feedback features. My goal was to also try to fix navigation issues mentioned in Assignment 2, Part C, to align with Nielsen's usability principles (Lecture 10). Despite the praise, the experiment only tested 6 participants registering for two courses: a very small sample size in courses and students.

Doing the experiment multiple times with multiple interfaces (more independent variables), or a larger sample size of participants could strengthen my findings. The opinions of six students did not represent all of the students at Carleton, not to mention that all of the participants were undergraduate students. Overall, I have reason to believe that this experiment was a success. My Figma redesign largely improves efficiency and usability over the original Carleton Central interface, when putting opinions, data and my observations from the experiment together.

Appendix A

Below is the anonymized raw data from the six students, with the dependent measures I used to measure the participants. Each student is labelled P1 to P6 (where P = participant). Note that NASA-TLX scores are rounded to 2 decimal places.

Raw Data Table 1 – Table with six students and all the dependent measures

Participant	Age	Interface	Time (seconds)	Errors	NASA-TLX scores
P1	22	Carleton Central	183	2	65.83
		Figma mockup	76	0	33.33
P2	21	Carleton Central	200	2	61.67
		Figma mockup	90	0	36.67
P3	21	Carleton Central	199	4	71.67
		Figma mockup	88	0	38.33
P4	20	Carleton Central	220	3	71.67
		Figma mockup	79	1	40.00
P5	23	Carleton Central	180	2	58.33
		Figma mockup	55	0	25.83
P6	21	Carleton Central	250	1	75.83
		Figma mockup	103	0	45.00

These values reflected the performance and data of all 6 students.

Below are the scores for each question by each student. The average is the sum of the scores of each question, where two questions were adjusted for reverse coding. There is no need to adjust any of the scores, because I did so already. The sum is each number displayed in the table divided by 6, since there were six questions asked.

Raw Data Table 2 – Table with six students and their question data for each question in the NASA-TLX survey. These scores are for the **Original Carleton Interface**.

Participant	Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Mean (Average)
P1	83	Reverse coded as 30	Reverse coded as 72	74	60	76	65.83
P2	70	Reverse coded as 28	Reverse coded as 80	68	71	53	61.67
P3	85	Reverse coded as 25	Reverse coded as 83	70	80	87	71.67
P4	90	Reverse coded as 29	Reverse coded as 88	59	77	87	71.67
P5	71	Reverse coded as 25	Reverse coded as 68	69	52	65	58.33
P6	92	Reverse coded as 33	Reverse coded as 91	83	67	89	75.83

Raw Data Table 3 – Table with six students and their question data for each question in the NASA-TLX survey. These scores are for the **Figma Mockup Interface**.

Participant	Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Mean (Average)
P1	25	Reverse coded as 18	Reverse coded as 18	92	20	27	33.33
P2	29	Reverse coded as 20	Reverse coded as 33	88	31	19	36.67
P3	30	Reverse coded as 22	Reverse coded as 26	95	36	21	38.33
P4	31	Reverse coded as 28	Reverse coded as 35	90	33	23	40.00
P5	16	Reverse coded as 10	Reverse coded as 15	93	10	11	25.83
P6	40	Reverse coded as 31	Reverse coded as 38	99	32	30	45.00

Appendix B

Here are the links to the various Qualtrics surveys/questionnaires, in order of completion:

- Consent form: https://carletonu.az1.qualtrics.com/jfe/form/SV_25JKDjPLza6A0dg
- NASA-TLX: https://carletonu.az1.qualtrics.com/jfe/form/SV_bdRjJDYap3iN5J4
- Post-Experiment Questionnaire:
https://carletonu.az1.qualtrics.com/jfe/form/SV_0xeVtAJB2OzHVRA
- Also, here is my Figma mockup interface:
<https://www.figma.com/design/wqUzYnnLBPrBniE2n14mqX/Simplifying-Carleton-Central-Course-Registration?node-id=0-1&t=2IVtNalKD86pYBqw-1>

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